Infosafe No™ LQAZU	Issue Date :April 2022	ISSUED by NUFARMAT

Product Name **Nufarm 2,4-D Amine 625 Herbicide**

Classified as hazardous

Section 1 - Identification

Product Identifier	Nufarm 2,4-D Amine 625 Herbicide
Company Product Codes / Numbers / Unique Identifiers	0478, NUL0384
Product Type	Group 1 Herbicide
Company Name	NUFARM AUSTRALIA LIMITED. (ABN 80 004 377 780)
Address	103-105 Pipe Road Laverton North Victoria 3026 AUSTRALIA
Telephone/Fax Number	Tel: +61 3 9282-1000
Emergency Phone Number	1800 033 498 (24hr Australia)
E-mail Address	SDSANZ@nufarm.com
Recommended use of the chemical and restrictions on use	For the control of broadleaf weeds in fallow before direct drilling or sowing of cereals and pastures; and in cereal crops, pastures, sugar cane, peanuts and non-agricultural areas as per the Directions of Use: table on the label. This is a PHENOXY HERBICIDE that can cause severe damage to susceptible crops such as cotton, grapes, tomatoes, oilseed crops and ornamentals.
Additional Information	This Safety Data Sheet describes the properties of the concentrated product. The physical properties and the assessments may not apply to the properties of the product once it has been diluted for application.

Section 2 - Hazard(s) Identification

GHS Classification of the Substance/Mixture	Classified as Hazardous according to the Globally Harmonised System of Classification and labelling of Chemicals (GHS) including Work, Health and Safety regulations, Australia. Not classified as Dangerous Goods according to the Australian Code for the Transport of Dangerous Goods by Road and Rail. (7th edition) Acute toxicity: Category 4 - Oral Sensitisation - skin: Category 1 Specific target organ toxicity (single exposure): Category 3 (Respiratory tract irritation) Eye damage/irritation: Category 1 Hazardous to the Aquatic Environment - Long-Term Hazard: Category 3
Signal Word	DANGER
Hazard Statement (s)	H302 Harmful if swallowed. H317 May cause an allergic skin reaction. H318 Causes serious eye damage. H335 May cause respiratory irritation. H412 Harmful to aquatic life with long lasting effects.
Pictogram (s)	Exclamation mark,Corrosion

Precautionary Statement – Prevention	P261 Avoid breathing dust/fume/gas/mist/vapours/spray. P270 Do not eat, drink or smoke when using this product. P271 Use only outdoors or in a well-ventilated area. P272 Contaminated work clothing should not be allowed out of the workplace. P273 Avoid release to the environment. P280(f) Wear protective gloves/protective clothing/eye protection/face protection.
Precautionary Statement – Response	P301+P312 IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell. P302+P352 IF ON SKIN: Wash with plenty of water. P330 Rinse mouth.

	P333+P313 If skin irritation or rash occurs: Get medical advice/attention. P362+P364 Take off contaminated clothing and wash it before reuse. P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing. P312 Call a POISON CENTER/doctor if you feel unwell. P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P310 Immediately call a POISON CENTER/doctor.
Precautionary Statement – Storage	P403+P233 Store in a well-ventilated place. Keep container tightly closed. P405 Store locked up.
Precautionary Statement – Disposal	P501 Dispose of contents/container to an approved waste disposal plant.

Section 3 - Composition and Information on Ingredients

Ingredients	Name	CAS	Proportion
	2,4-D present as dimethylamine and diethanolamine salts		40-50 %
	Ingredients determined not to be hazardous, including water		Balance

Section 4 - First Aid Measures

Inhalation	If inhaled, remove affected person from contaminated area. Apply artificial respiration if not breathing. Seek medical attention.
Ingestion	Do not induce vomiting. Wash out mouth thoroughly with water. Seek immediate medical attention.
Skin	Remove all contaminated clothing immediately. Wash affected area thoroughly with soap and water. Wash contaminated clothing before reuse or discard. Seek medical attention.
Eye	If in eyes, hold eyelids apart and flush the eyes continuously with running water. Remove contact lenses. Continue flushing until advised to stop by the Poisons Information Centre or a doctor, or for at least 15 minutes. Seek immediate medical attention.
First Aid Facilities	Eye wash, safety shower and normal washroom facilities.
Advice to Doctor	Treat symptomatically.
Other Information	For advice in an emergency, contact a Poisons Information Centre (Phone Australia 131 126) or a doctor at once.

Section 5 - Firefighting Measures

Suitable Extinguishing Media	Use appropriate fire extinguisher for surrounding environment.
Hazards from Combustion Products	Under fire conditions this product may emit toxic and/or irritating fumes and gases including hydrogen chloride, phosgene, carbon monoxide and carbon dioxide.
Specific Hazards Arising from the Chemical	This product is non combustible.
Decomposition Temperature	Not available
Precautions in connection with Fire	Fire fighters should wear full protective clothing and self-contained breathing apparatus (SCBA) operated in positive pressure mode. Fight fire from safe location.

Section 6 - Accidental Release Measures

Emergency Procedures	Wear appropriate personal protective equipment and clothing to prevent exposure. Increase ventilation. If possible contain the spill. Place inert absorbent material onto spillage. Collect the material and place into a suitable labelled container. Do not dilute material but contain. Dispose of waste according to the applicable local and national regulations. If

	contamination of sewers or waterways occurs inform the local water and waste management authorities in accordance with local regulations.
Environmental Precautions	This product is a herbicide and spills can damage crops, pastures and desirable vegetation.

Section 7 - Handling and Storage

Precautions for Safe Handling	Avoid inhalation of vapours and mists, and skin or eye contact. Use only in a well ventilated area. Keep containers sealed when not in use. Prevent the build up of mists or vapours in the work atmosphere. Maintain high standards of personal hygiene i.e. Washing hands prior to eating, drinking, smoking or using toilet facilities.
Conditions for safe storage, including any incompatibilities	Store in a cool, dry, well-ventilated area, out of direct sunlight. Store in suitable, labelled containers. Keep containers tightly closed. Store away from incompatible materials. Ensure that storage conditions comply with applicable local and national regulations. Protect from freezing.
Other Information	Equipment that has been used for this chemical should not be used for the application of other materials to sensitive plants, unless it has been well washed out with hot soapy water or 1% solution of ammonia, followed by several clear water rinse.

Section 8 - Exposure Controls and Personal Protection

Occupational Exposure Limit (OEL) Values	No exposure standards have been established for the mixture. Exposure limits for ingredients and dimethylamine which could be formed on rection is given below: 2,4-Dichlorophenoxyacetic acid TWA: 10 mg/m³ Note: Sen Dimethylamine TWA: 2 ppm, 3.8 mg/m³ STEL: 6 ppm, 11 mg/m³ TWA (Time Weighted Average): The average airborne concentration of a particular substance when calculated over a normal eight-hour working day, for a five-day week. STEL (Short Term Exposure Limit): The average airborne concentration over a 15 minute period which should not be exceeded at any time during a normal eight-hour workday. 'Sen' Notice: The substance may cause sensitisation by skin contact or by inhalation. Source: Safe Work Australia.
Biological Limit Values (BLV)	No biological limits allocated.
Control Banding	Not available
Engineering Controls	This substance is hazardous and should be used with a local exhaust ventilation system, drawing vapours away from workers' breathing zone. If the engineering controls are not sufficient to maintain concentrations of vapours/mists below the exposure standards, suitable respiratory protection must be worn.
Respiratory Protection	If engineering controls are not effective in controlling airborne exposure then an approved respirator with a replaceable vapor/mist filter should be used. Refer to relevant regulations for further information concerning respiratory protective requirements. Reference should be made to Australian Standards AS/NZS 1715, Selection, Use and Maintenance of Respiratory Protective Devices; and AS/NZS 1716, Respiratory Protective Devices, in order to make any necessary changes for individual circumstances.
Eye and Face Protection	Safety glasses with full face shield should be used. Eye protection devices should conform to relevant regulations. Eye protection should conform with Australian/New Zealand Standard AS/NZS 1337 (series) - Eye Protectors for Industrial Applications.
Hand Protection	Wear gloves of impervious material such as elbow length PVC gloves. Final choice of appropriate gloves will vary according to individual circumstances. i.e. methods of handling or according to risk assessments undertaken.

	Occupational protective gloves should conform to relevant regulations. Reference should be made to AS/NZS 2161.1: Occupational protective gloves - Selection, use and maintenance.
Thermal Hazards	No further relevant information available.
Body Protection	Suitable protective work wear, e.g. cotton overalls buttoned at neck and wrist is recommended. Chemical resistant apron is recommended where large quantities are handled. When preparing spray, wear PVC or rubber apron.
Hygiene Measures	Do not eat, drink or smoke when using this product.
Requirements Concerning Special Training	People who use pesticides in their job or business should have training in the application of the materials.

Section 9 - Physical and Chemical Properties

Form	Liquid
Appearance	Clear liquid
Colour	Red-brown liquid
Odour	Ammoniacal odour.
Melting Point	<0°C
Boiling Point	>100°C (for water)
Decomposition Temperature	Not available
Solubility in Water	Soluble in water.
Specific Gravity	1.254
pH	8.5-9.5
Vapour Pressure	Not available
Relative Vapour Density (Air=1)	Not available
Evaporation Rate	Not available
Odour Threshold	Not available
Viscosity	Not available
Volatile Component	30%
Partition Coefficient: n-octanol/water (log value)	Not available
Flash Point	Not applicable
Flammability	Non combustible
Auto-ignition Temperature	Not available
Flammable Limits - Lower	Not available
Flammable Limits - Upper	Not available
Explosion Properties	Not available
Oxidising Properties	Not available

Section 10 - Stability and Reactivity

Chemical Stability	Stable under normal conditions of handling and storage.
Possibility of Hazardous Reactions	Not available

Conditions to Avoid	Extremes of temperature and direct sunlight.
Incompatible Materials	Strong oxidising agents. Reaction of the concentrate or spray mix with acids will precipitate solid 2,4-D acid and largely de-activate the product and cause blockages in spray equipment. The addition of a strong alkali such as caustic soda will cause release of dimethylamine vapour. Dimethylamine is moderately toxic, LD50 (oral, rat) is 700 mg/kg.
Hazardous Decomposition Products	Thermal decomposition may result in the release of toxic and/or irritating fumes, smoke and gases including: hydrogen chloride, phosgene, carbon monoxide and carbon dioxide.
Reactivity and Stability	Reacts with incompatible materials.
Hazardous Polymerization	Will not occur.

Section 11 - Toxicological Information

Toxicology Information	Toxicity data is given below.
Acute Toxicity - Oral	LD50 (rat): 699 mg/kg for 2,4-dichlorphenoxyacetic acid
Acute Toxicity - Dermal	LD50 (rabbit): >2000 mg/kg
Acute Toxicity - Inhalation	LC50 (rat): >1.79 mg/l/4h
Ingestion	Harmful if swallowed. Ingestion of the product in relatively large amounts can result in headache, nausea, lethargy, motor weakness and incoordination.
Inhalation	May cause respiratory irritation. The components of the product are of low volatility and no adverse effects are expected from handling the concentrate. A moderate hazard exists from inhalation of spray and care should be taken to avoid inhalation of spray mists.
Skin	May be irritating to skin. The symptoms may include redness, itching and swelling. Prolonged contact of the concentrate with skin will result in absorption of some 2,4-D which can be harmful. May cause an allergic skin reaction.
Eye	Causes serious eye damage. Eye contact will cause stinging, blurring, tearing, severe pain and possible burns, necrosis, permanent damage and blindness. Prolonged contact with the concentrate may cause damage to the eye.
Respiratory Sensitisation	Not expected to be a respiratory sensitiser.
Skin Sensitisation	May cause an allergic skin reaction.
Germ Cell Mutagenicity	Not considered to be a mutagenic hazard.
Carcinogenicity	Not considered to be a carcinogenic hazard. The weight of the evidence is that 2,4-D is not carcinogenic.
Reproductive Toxicity	Not considered to be toxic to reproduction.
STOT - Single Exposure	May cause respiratory irritation.
STOT - Repeated Exposure	Not expected to cause toxicity to a specific target organ. Repeated absorption of relatively large amounts of 2,4-D presents a risk to the liver and kidneys.
Aspiration Hazard	Not expected to be an aspiration hazard.
Other Information	The Australian Acceptable Daily Intake (ADI) for 2,4-D for a human is 0.01 mg/kg/day, set for the public for daily, lifetime exposure. This is based on the NOEL of 1 mg/kg/day, the level determined to show no effects during long term exposure for the most sensitive indicators and the most sensitive species. (Ref: Australian Pesticides and Veterinary Medicines Authority (APVMA) December 2021).

Section 12 - Ecological Information

Ecotoxicity	Harmful to aquatic life with long lasting effects. 2,4-D amine products do not appear to pose any threat to birds. 2,4-D amine products do not appear to pose any threat to fish or other aquatic organisms other than in very high concentrations. Not toxic to fish, birds and bees.
Persistence and Degradability	Half life in soil is typically 7 days. Loss from soil is principally by microbial degradation. Not inhibitory in sewage system, 2,4-D is rapidly biodegraded.
Mobility	Rapid degradation in soil prevents significant downward movement under normal conditions.
Bioaccumulative Potential	Not available
Other Adverse Effects	Spray drift can cause damage, read the label for more information.
Environmental Protection	Prevent this material entering waterways, drains and sewers.
Acute Toxicity - Fish	LC50 (rainbow trout): approx. 100 mg/l/96h. LC50 (fish): 184 mg/l/48h
Acute Toxicity - Other Organisms	LD50 (mallard ducks): >1000 mg/kg LD50 (bees): 104.5 µg/bee.
Hazardous to the Ozone Layer	This product is not expected to deplete the ozone layer.

Section 13 - Disposal Considerations

Disposal Considerations	Dispose of waste according to applicable local and national regulations. Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes including emptied containers are controlled wastes and should be disposed of in accordance with all applicable local and national regulations. To minimise personal exposure to the chemical, refer to Section 8 — Exposure controls and personal protection.
Container Disposal and Methods	Do not use this container for any other purpose. Triple or preferably pressure rinse containers before disposal. Add rinsings to the spray tank. If recycling, replace cap and return clean containers to recycler or designated collection point. If not recycling, puncture or shred and bury containers in local authority landfill. If no landfill is available, bury the containers below 500mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt.

Section 14 - Transport Information

Transport Information	Not classified as Dangerous Goods according to the Australian Code for the Transport of Dangerous Goods by Road and Rail. (7th edition) Not classified as Dangerous Goods by the criteria of the International Air Transport Association (IATA) Dangerous Goods Regulations for transport by air. Not classified as Dangerous Goods by the criteria of the International Maritime Dangerous Goods Code (IMDG Code) for transport by sea.
Special Precautions for Use	Not available
IMDG Marine Pollutant	No
Transport in Bulk	Not available

Section 15 - Regulatory Information

Regulatory Information	Classified as Hazardous according to the Globally Harmonised System of Classification and Labelling of Chemicals (GHS) including Work, Health and Safety Regulations, Australia. Classified as a Scheduled Poison according to the Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP).
Poisons Schedule	S6

Montreal Protocol	Not listed
Stockholm Convention	Not listed
Rotterdam Convention	Not listed
International Convention for the Prevention of Pollution from Ships (MARPOL)	Not available
Agricultural and Veterinary Chemicals Act 1994	This product has been registered with the Australian Pesticides and Veterinary Medicines Authority (APVMA) by Nufarm under the number 56763.
Basel Convention	Not available

Section 16 - Any Other Relevant Information

Date of Preparation	SDS Reviewed: 14 April 2022
Version Number	2.0
Revisions Made	The SDS was reviewed. Minor changes were made to the information and product has been rebranded.
Literature References	Preparation of Safety Data Sheets for Hazardous Chemicals Code of Practice. Standard for the Uniform Scheduling of Medicines and Poisons. Australian Code for the Transport of Dangerous Goods by Road & Rail. Work Health and Safety Regulations, Schedule 10: Prohibited carcinogens, restricted carcinogens and restricted hazardous chemicals. Code of Practice for Supply Diversion into Illicit Drug Manufacture. National Code of Practice for Chemicals of Security Concern. Agricultural Compounds and Veterinary Chemicals Act. International Agency for Research on Cancer (IARC) Monographs. Montreal Protocol on Substances that Deplete the Ozone Layer. Stockholm Convention on Persistent Organic Pollutants (POPs). Rotterdam Convention on the Prior Informed Consent Procedure for Certain Hazardous Chemicals and Pesticides in International Trade. Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal. International Air Transport Association (IATA) Dangerous Goods Regulations. International Maritime Dangerous Goods (IMDG) Code. Workplace exposure standards for airborne contaminants. Adopted biological exposure determinants, American Conference of Industrial Hygienists (ACGIH). Globally Harmonised System of Classification and Labelling of Chemicals (7th revised edition). Code of Practice: Managing Noise and Preventing Hearing Loss at Work.

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